Fibertec environmental services

Thursday, October 27, 2022

Fibertec Project Number: A11477

Project Identification: NPDES- Semi Annual PFAS /

Submittal Date: 10/13/2022

Mrs. Cindy Tomaszewski
City of Cadillac
200 Lake Street
Cadillac, MI 49601

Dear Mrs. Tomaszewski,

Thank you for selecting Fibertec Environmental Services as your analytical laboratory. The samples you submitted have been analyzed in accordance with NELAC standards and the results compiled in the attached report. Any exceptions to NELAC compliance are noted in the report. These results apply only to those samples submitted. Please note TO-15 samples will be disposed of 7 calendar days after the reporting date. All other samples will be disposed of 30 days after the reporting date.

If you have any questions regarding these results or if we may be of further assistance to you, please contact me at (517) 699-0345.

Sincerely,

Bailey Welch

By Bailey Welch at 2:32 PM, Oct 27, 2022

For Daryl P. Strandbergh
Laboratory Director

Enclosures

Analytical Laboratory Report
Laboratory Project Number: A11477
Laboratory Sample Number: A11477-001

Order: A11477
Date: 10/27/22

Client Identification:	**City of Cadillac**	Sample Description:	**Field Blank**	Chain of Custody:	**209192**
Client Project Name:	**NPDES- Semi Annual PFAS**	Sample No:		Collect Date:	**10/13/22**
Client Project No:	**NA**	Sample Matrix:	**Blank: Field**	Collect Time:	**11:01**

Sample Comments:

Definitions: Q: Qualifier (see definitions at end of report) NA: Not Applicable ‡: Parameter not included in NELAC Scope of Analysis.

PFAS **Aliquot ID: A11477-001** **Matrix: Blank: Field**
Method: EPA 0537.1 (Modified) **Description: Field Blank**

Parameter(s)	Result	Q	Units	Reporting Limit	Dilution	Preparation P. Date	Preparation P. Batch	Analysis A. Date	Analysis A. Batch	Init.
‡ 1. ADONA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 2. 9Cl-PF3ONS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 3. 11Cl-PF3OUdS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 4. N-EtFOSAA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 5. FtS 4:2	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 6. FtS 6:2	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 7. FtS 8:2	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 8. HFPO-DA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 9. N-MeFOSAA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 10. PFBA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 11. PFBS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 12. PFBSA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 13. PFDA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 14. PFDoA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 15. PFDS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 16. PFECHS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 17. PFHpA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 18. PFHpS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 19. PFHxA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 20. PFHxSA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 21. PFHxS-Total	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 22. PFNA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 23. PFNS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 24. PFOA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 25. PFOSA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 26. PFOS-Total	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 27. PFPeA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 28. PFPeS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 29. PFTeA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 30. PFTriA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 31. PFUnA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG

Fibertec environmental services

Analytical Laboratory Report
Laboratory Project Number: A11477
Laboratory Sample Number: A11477-002

Order: A11477
Date: 10/27/22

Client Identification:	**City of Cadillac**	Sample Description:	**Effluent**	Chain of Custody:	**209192**
Client Project Name:	**NPDES- Semi Annual PFAS**	Sample No:		Collect Date:	**10/13/22**
Client Project No:	**NA**	Sample Matrix:	**Wastewater**	Collect Time:	**11:07**

Sample Comments:

Definitions: Q: Qualifier (see definitions at end of report) NA: Not Applicable ‡: Parameter not included in NELAC Scope of Analysis.

PFAS **Aliquot ID: A11477-002** **Matrix: Wastewater**
Method: EPA 0537.1 (Modified) **Description: Effluent**

						Preparation		Analysis		
Parameter(s)	Result	Q	Units	Reporting Limit	Dilution	P. Date	P. Batch	A. Date	A. Batch	Init.
‡ 1. ADONA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 2. 9Cl-PF3ONS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 3. 11Cl-PF3OUdS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 4. N-EtFOSAA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 5. FtS 4:2	U	EIS+	ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 6. FtS 6:2	U	EIS+	ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 7. FtS 8:2	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 8. HFPO-DA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 9. N-MeFOSAA	**1.6**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 10. PFBA	**4.6**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 11. PFBS	**34**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 12. PFBSA	**26**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 13. PFDA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 14. PFDoA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 15. PFDS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 16. PFECHS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 17. PFHpA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 18. PFHpS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 19. PFHxA	**41**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 20. PFHxSA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 21. PFHxS-Total	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 22. PFNA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 23. PFNS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 24. PFOA	**1.6**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 25. PFOSA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 26. PFOS-Total	**1.4**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 27. PFPeA	**3.1**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 28. PFPeS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 29. PFTeA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 30. PFTriA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡ 31. PFUnA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG

Analytical Laboratory Report
Laboratory Project Number: A11477
Laboratory Sample Number: A11477-003

Order: A11477
Date: 10/27/22

Client Identification:	**City of Cadillac**	Sample Description:	**Effluent Duplicate**	Chain of Custody:	**209192**
Client Project Name:	**NPDES- Semi Annual PFAS**	Sample No:		Collect Date:	**10/13/22**
Client Project No:	**NA**	Sample Matrix:	**Wastewater**	Collect Time:	**11:14**

Sample Comments:

Definitions: Q: Qualifier (see definitions at end of report) NA: Not Applicable ‡: Parameter not included in NELAC Scope of Analysis.

PFAS **Aliquot ID: A11477-003** **Matrix: Wastewater**
Method: EPA 0537.1 (Modified) **Description: Effluent Duplicate**

	Parameter(s)	Result	Q	Units	Reporting Limit	Dilution	Preparation		Analysis		
							P. Date	P. Batch	A. Date	A. Batch	Init.
‡	1. ADONA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	2. 9Cl-PF3ONS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	3. 11Cl-PF3OUdS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	4. N-EtFOSAA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	5. FtS 4:2	U	EIS+	ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	6. FtS 6:2	U	EIS+	ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	7. FtS 8:2	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	8. HFPO-DA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	9. N-MeFOSAA	**1.5**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	10. PFBA	**4.6**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	11. PFBS	**36**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	12. PFBSA	**27**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	13. PFDA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	14. PFDoA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	15. PFDS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	16. PFECHS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	17. PFHpA	**1.1**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	18. PFHpS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	19. PFHxA	**43**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	20. PFHxSA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	21. PFHxS-Total	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	22. PFNA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	23. PFNS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	24. PFOA	**1.7**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	25. PFOSA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	26. PFOS-Total	**1.4**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	27. PFPeA	**3.3**		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	28. PFPeS	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	29. PFTeA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	30. PFTriA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG
‡	31. PFUnA	U		ng/L	1.0	1.0	10/24/22	PS22J24G	10/26/22	SM22J26A	SKG

Acronym (Param)	Analyte Name	CAS Number
1. ADONA	4,8-dioxa-3H-perfluorononanoic acid	919005-14-4
2. 9Cl-PF3ONS	9-chlorohexadecafluoro-3-oxanone-1-sulfonic acid	756426-58-1
3. 11Cl-PF3OUdS	11-chloroeicosafluoro-3-oxaundecane-1-sulfonic acid	763051-92-9
4. N-EtFOSAA	2-(N-Ethylperfluorooctanesulfonamido) acetic acid	2991-50-6
5. FtS 4:2	Fluorotelomer sulphonic acid 4:2	757124-72-4
6. FtS 6:2	Fluorotelomer sulphonic acid 6:2	27619-97-2
7. FtS 8:2	Fluorotelomer sulphonic acid 8:2	39108-34-4
8. HFPO-DA	Hexafluoropropylene oxide dimer acid	13252-13-6
9. N-MeFOSAA	2-(N-Methylperfluorooctanesulfonamido) acetic acid	2355-31-9
10. PFBA	Perfluorobutanoic acid	375-22-4
11. PFBS	Perfluorobutanesulfonic acid	375-73-5
12. PFBSA	PFBSA	30334-69-1
13. PFDA	Perfluorodecanoic acid	335-76-2
14. PFDoA	Perfluorododecanoic acid	307-55-1
15. PFDS	Perfluorodecanesulfonic acid	335-77-3
16. PFECHS	PFECHS	335-24-0
17. PFHpA	Perfluoroheptanoic acid	375-85-9
18. PFHpS	Perfluoroheptanesulfonic acid	375-92-8
19. PFHxA	Perfluorohexanoic acid	307-24-4
20. PFHxSA	PFHxSA	41997-13-1
21. PFHxS-Total	Perfluorohexanesulfonic acid	355-46-4
22. PFNA	Perfluorononanoic acid	375-95-1
23. PFNS	Perfluorononanesulfonic acid	68259-12-1
24. PFOA	Perfluorooctanoic acid	335-67-1
25. PFOSA	Perfluorooctanesulfonamide	754-91-6
26. PFOS-Total	Perfluorooctanesulfonic acid	1763-23-1
27. PFPeA	Perfluoropentanoic acid	2706-90-3
28. PFPeS	Perfluoropentanesulfonic acid	2706-91-4
29. PFTeA	Perfluorotetradecanoic acid	376-06-7
30. PFTriA	Perfluorotridecanoic acid	72629-94-8
31. PFUnA	Perfluoroundecanoic acid	2058-94-8

Definitions/ Qualifiers:

A: Spike recovery or precision unusable due to dilution.
B: The analyte was detected in the associated method blank.
E: The analyte was detected at a concentration greater than the calibration range, therefore the result is estimated.
J: The concentration is an estimated value.
M: Modified Method
U: The analyte was not detected at or above the reporting limit.
X: Matrix Interference has resulted in a raised reporting limit or distorted result.
W: Results reported on a wet-weight basis.
***:** Value reported is outside QC limits

Exception Summary:

EIS+ : The Isotope Dilution/Extracted Internal Standard area exceeds the upper control limit.

Analysis Locations:

All analyses performed in Holt.

Accreditation Number(s):

T104704518-22-14 (TX)

Fibertec environmental services

Analytical Laboratory
1914 Holloway Drive, Holt, MI 48842
Phone: 517 699 0345
Fax: 517 699 0388
email: lab@fibertec.us

8660 S. Mackinaw Trail, Cadillac, MI 49601
Phone: 231 775 8368
Fax: 231 775 8584

Geoprobe
11766 E. Grand River Rd., Brighton, MI 48116
Phone: 810 220 3300
Fax: 810 220 3311

Chain of Custody # 209192

Client Name: CITY OF CADILLAC WWTP
Contact Person: CINDY TOMASZEWSKI
Project Name/ Number: NPDES - SEMIANNUAL PFAS
Email distribution list: lab@cadillac-mi.net
Quote#:
Purchase Order#:

Matrix Code: S Soil; A Air; O Oil; P Wipe; GW Ground Water; SW Surface Water; WW Waste Water; X Other: Specify — PFAS Free H_2O

Deliverables: Level 2 [ ]; Level 3 [ ]; Level 4 [ ]; EDD [X] — MI WATERS (NPDES)

Date	Time	Sample #	Client Sample Descriptor	MATRIX (SEE RIGHT CORNER FOR CODE)	# OF CONTAINERS	PFAS (EPA 537.1 M)	HOLD SAMPLE	Remarks
10-13	11:01		Field Blank	X	1	X		
10-13	11:07		Effluent	WW	3	X		
10-13	11:14		Effluent Duplicate	WW	3	X		

Received By Lab
OCT 13 2022
Initials: JS

Received On Ice

Comments: Any Questions - Just Call. Thx. CAT

	Date/ Time	
Sampled/Relinquished By: Chris Marquar	10-13-22 1245	Received By: E. Silvereth Griswold
Relinquished By: E. Silvereth Griswold	10-13-22 14:30	Received By: Lori Allen
Relinquished By: Lori Allen	10/13/22 17:16	Received By Laboratory: [signature]

Turnaround Time ALL RESULTS WILL BE SENT BY THE END OF THE BUSINESS DAY

____ 1 bus. day ____ 2 bus. days ____ 3 bus. days ____ 4 bus. days

X ~~5-7~~ 10-14? bus. days (standard) PFAS   Other (specify time/date requirement): ____________

LAB USE ONLY
Fibertec project number: A11477
Temperature upon receipt at Lab: 4.2°C

Please see back for terms and conditions